ECON2050 Tutorial 10

Optimization III: Nonlinear Programming

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Today's Plan

- Assignment 3 has been released.
- Due date: 30/10/2026 4:00 pm.
- I'll go through some important points for the assignment at next week tutorial.
- Tutorial questions: Q1–Q3

Tutorial Q1

Question 1

Consider the maximization problem

$$\max_{(x_1,x_2)\in\mathbb{R}^2} -(x_1-4)^2-(x_2-4)^2$$

subject to

$$x_1+x_2\leq 4, \qquad x_1+3x_2\leq 9.$$

(a) Are the Lagrange conditions necessary?

(b) Write down the Lagrange conditions.

(c) Find all points that satisfy the Lagrange conditions.

Recap: Lagrange conditions for inequalities

Consider

$$\max_{x \in S} f(x) \quad \text{s.t.} \quad g_i(x) \leq 0 \quad (i = 1, \ldots, m).$$

Lagrange conditions

At a candidate optimum x^*, seek multipliers $\lambda^* = (\lambda_1^*, \ldots, \lambda_m^*)$ satisfying

$$\nabla f(x^*) - \sum_{i=1}^{m} \lambda_i^* \nabla g_i(x^*) = 0.$$

$$\begin{aligned} g_i(x^*) &\leq 0 \quad \text{(primal feasibility)}, \\ \lambda_i^* &\geq 0 \quad \text{(multiplier signs)}, \\ \lambda_i^* g_i(x^*) &= 0 \quad \text{(complementary slackness)}, \end{aligned}$$

for every $i = 1, \ldots, m$.

Recap: The Lagrangian

Lagrangian function

For the inequality-constrained maximization problem, define

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}) = f(\boldsymbol{x}) - \sum_{i=1}^{m} \lambda_i g_i(\boldsymbol{x}).$$

Stationarity

The first part of the Lagrange conditions can be written as

$$\boldsymbol{\nabla}_{\boldsymbol{x}} \mathcal{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*) = \mathbf{0}.$$

Stationarity must be accompanied by feasibility, multiplier sign restrictions, and complementary slackness.

Recap: When are Lagrange conditions necessary?

Affine constraint functions

If the continuously differentiable objective f is maximized subject to *affine* inequalities $g_i(\boldsymbol{x}) \leq 0$, then every solution satisfies the Lagrange conditions for some non-negative multiplier vector $\boldsymbol{\lambda}^*$.

Nonlinear constraint functions

For continuously differentiable constraints, the Lagrange conditions are also necessary if the gradients of the *binding* constraint functions at the solution are linearly independent.

Recap: Using complementary slackness

An approach to solving Lagrange conditions

1. Write each inequality as $g_i(\boldsymbol{x}) \leq 0$.
2. Write stationarity, feasibility, multiplier signs, and $\lambda_i g_i(\boldsymbol{x}) = 0$.
3. For each inequality, consider the cases $\lambda_i = 0$ and $\lambda_i > 0$.
4. If $\lambda_i > 0$, complementary slackness gives $g_i(\boldsymbol{x}) = 0$.
5. Retain only candidates satisfying *all* conditions.

Tutorial Q1: Knowledge used

Knowledge used

- **Necessary conditions**: with affine constraint functions, every optimizer must satisfy the Lagrange conditions.
- **Inequality constraints**: write $g_i(x_1, x_2) \leq 0$, with $\lambda_i \geq 0$.
- **Stationarity**: $\boldsymbol{\nabla} f - \lambda_1 \boldsymbol{\nabla} g_1 - \lambda_2 \boldsymbol{\nabla} g_2 = \mathbf{0}$.
- **Complementary slackness**: $\lambda_i g_i = 0$.
- Analyze whether each multiplier is zero or strictly positive.

Tutorial Q1(a)–(b): Lagrange conditions

Solution

(a) The constraint functions

$$g_1(x_1, x_2) = x_1 + x_2 - 4, \qquad g_2(x_1, x_2) = x_1 + 3x_2 - 9$$

are affine. Hence the Lagrange conditions are necessary.

(b) Stationarity, feasibility, multiplier signs, and complementary slackness give

$$\begin{aligned} -2(x_1 - 4) - \lambda_1 - \lambda_2 &= 0, \\ -2(x_2 - 4) - \lambda_1 - 3\lambda_2 &= 0, \\ x_1 + x_2 &\le 4, \qquad \lambda_1 \ge 0, \\ x_1 + 3x_2 &\le 9, \qquad \lambda_2 \ge 0, \\ \lambda_1(x_1 + x_2 - 4) &= 0, \\ \lambda_2(x_1 + 3x_2 - 9) &= 0. \end{aligned}$$

Tutorial Q1(c): Cases 1 and 2

Solution

Case 1: $\lambda_1 = \lambda_2 = 0$.
Stationarity gives $x_1 = x_2 = 4$, but

$$x_1 + x_2 = 8 > 4.$$

This case is infeasible.

Case 2: $\lambda_1 > 0$, $\lambda_2 = 0$.
Stationarity yields

$$x_1 = x_2 = 4 - \frac{\lambda_1}{2}.$$

Complementary slackness requires $x_1 + x_2 = 4$, so

$$x_1 = x_2 = 2, \qquad \lambda_1 = 4, \qquad \lambda_2 = 0.$$

Since $2 + 3(2) = 8 \leq 9$, this case is valid.

Tutorial Q1(c): Case 3

Solution

Case 3: $\lambda_1 = 0$, $\lambda_2 > 0$.
The first stationarity equation implies

$$\lambda_2 = -2(x_1 - 4).$$

Substitution into the second gives

$$-2(x_2 - 4) + 6(x_1 - 4) = 0 \quad \Longleftrightarrow \quad 3x_1 - x_2 = 8.$$

As $\lambda_2 > 0$, the second constraint binds:

$$x_1 + 3x_2 = 9.$$

Solving yields

$$(x_1, x_2) = \left(\frac{33}{10}, \frac{19}{10}\right).$$

But $x_1 + x_2 = \frac{26}{5} > 4$; hence this case is infeasible.

Tutorial Q1(c): Case 4 and conclusion

Solution

Case 4: $\lambda_1 > 0$, $\lambda_2 > 0$.
Both constraints bind:

$$x_1 + x_2 = 4, \qquad x_1 + 3x_2 = 9.$$

Thus $(x_1, x_2) = (\frac{3}{2}, \frac{5}{2})$. Stationarity becomes

$$5 - \lambda_1 - \lambda_2 = 0, \qquad 3 - \lambda_1 - 3\lambda_2 = 0,$$

which gives $\lambda_1 = 6$, $\lambda_2 = -1$. Since $\lambda_2 < 0$, this case is ruled out.

The unique point satisfying the Lagrange conditions is

$$(x_1^*, x_2^*, \lambda_1^*, \lambda_2^*) = (2, 2, 4, 0).$$

The graphical check in the official solution confirms that $(2, 2)$ is the unique maximizer.

Tutorial Q2

Question 2

Consider the maximization problem

$$\max_{(x,y)\in\mathbb{R}^2} \; x - y \qquad \text{subject to} \qquad x^2 + y^2 = 8.$$

(a) Does a solution exist? Explain.

(b) Find all solutions to this maximization problem.

Recap: Extreme Value Theorem

Existence of global extrema

If $D \subseteq \mathbb{R}^n$ is compact and $f : D \to \mathbb{R}$ is continuous, then f has at least one global maximum and at least one global minimum on D.

Convexity of the constraint set

The constraint set of any linear program is convex because it is described by linear inequalities and equalities.

Recap: Constraint qualifications and candidates

Checking the gradients

- A convenient sufficient check is that the gradients of *all* constraints are linearly independent at every feasible point.
- If this fails, inspect where the gradients of the *binding* constraints are linearly dependent.

Important caution

Without the required assumptions, a genuine solution of a nonlinear program need not satisfy the Lagrange conditions. Even when they are necessary, points satisfying them are *candidates*, not automatically global solutions.

Recap: Equality constraints

Consider the maximization problem

$$\max_{x \in S} \; f(x) \qquad \text{s.t.} \qquad g_i(x) = 0, \quad i = 1, \ldots, m.$$

Lagrange conditions

If the constraint functions are affine, or their gradients are linearly independent at the solution, then every solution satisfies

$$\nabla f(x^*) - \sum_{i=1}^{m} \lambda_i^* \nabla g_i(x^*) = \mathbf{0},$$

$$g_i(x^*) = 0, \qquad i = 1, \ldots, m.$$

Here $\lambda^* \in \mathbb{R}^m$ is **unrestricted in sign**. There is no complementary slackness condition for equalities.

Tutorial Q2: Knowledge used

Knowledge used

- **Extreme Value Theorem**: a continuous objective function attains a maximum on a compact constraint set.
- **Equality-constrained Lagrange conditions**: multipliers may be negative.
- **Necessary condition**: if the gradient of the equality constraint is nonzero at every feasible point, the Lagrange conditions are necessary.
- Evaluate the objective at all candidate points to identify the global maximum.

Tutorial Q2(a): Existence

Solution

The objective function

$$f(x, y) = x - y$$

is linear and therefore continuous.

The constraint set is

$$C = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 8\},$$

the circle centered at the origin with radius $2\sqrt{2}$. It is closed and bounded, hence **compact**.

By the **Extreme Value Theorem**, the maximization problem has a solution.

Tutorial Q2(b): Necessary Lagrange conditions

Solution

Define the equality constraint

$$g(x, y) = x^2 + y^2 - 8.$$

Its gradient is

$$\nabla g(x, y) = (2x, 2y).$$

The gradient can be zero only at $(0, 0)$, which is not in the constraint set. Therefore, the equality constraint satisfies the linear-independence condition, and the Lagrange conditions are necessary.

They are

$$\begin{aligned} 1 - 2\lambda x &= 0, \\ -1 - 2\lambda y &= 0, \\ x^2 + y^2 - 8 &= 0, \end{aligned}$$

where λ is unrestricted in sign.

Tutorial Q2(b): Candidates and maximum

Solution

The first stationarity equation implies $\lambda \neq 0$. Together, the first two equations imply $y = -x$. Using the equality constraint,

$$x^2 + (-x)^2 = 8 \quad \Longrightarrow \quad x = \pm 2.$$

Thus there are two Lagrange candidates:

$$(x, y, \lambda) = \left(2, -2, \frac{1}{4}\right), \qquad \left(-2, 2, -\frac{1}{4}\right).$$

Evaluate the objective:

$$f(2, -2) = 4, \qquad f(-2, 2) = -4.$$

Since a maximum exists, and every maximizer must satisfy the Lagrange conditions, the unique solution is

$$(x^*, y^*) = (2, -2), \qquad f(x^*, y^*) = 4.$$

Tutorial Q3

Question 3

For $x > 0$, $y > 0$, $z > 0$, consider

$$\max \left(2 \ln x + 5 \ln y + \ln z\right) \qquad \text{s.t.} \qquad x^2 + y^2 + z = 32.$$

(a) Write down the Lagrange conditions, and find the unique point satisfying them.

(b) Assuming that the point in (a) is indeed the unique solution, approximate the optimal value when the constraint becomes

$$x^2 + y^2 + z = 32.1.$$

Do not solve the modified problem.

Recap: Economic interpretation of multipliers

Lagrange multipliers as dual variables

Lagrange multipliers inherit the interpretation of dual variables: they measure the marginal value of changing the corresponding resource or constraint level.

Small changes in a constraint bound

For a sufficiently small change Δb in a constraint bound b, the change in the optimized objective value can be approximated by

$$V(b + \Delta b) \approx V(b) + \lambda^* \Delta b.$$

For equality constraints the multiplier is unrestricted in sign.

Tutorial Q3: Knowledge used

Knowledge used

- **Equality constraints**: $g(x, y, z) = 0$, with an unrestricted Lagrange multiplier λ.
- **Necessary conditions**: the gradient of the binding equality constraint must be nonzero for the Lagrange conditions to be necessary.
- **Stationarity and feasibility**: solve $\nabla f - \lambda \nabla g = \mathbf{0}$ together with $g = 0$.
- **Multiplier interpretation**: a small change in the constraint bound changes the optimal value approximately by λ^* times the change in that bound.

Tutorial Q3(a): Lagrange conditions

Solution

Let

$$f(x, y, z) = 2 \ln x + 5 \ln y + \ln z, \qquad g(x, y, z) = x^2 + y^2 + z - 32.$$

Since

$$\nabla g(x, y, z) = (2x, 2y, 1) \neq \mathbf{0},$$

the equality-constraint Lagrange conditions are necessary.

They are

$$\begin{aligned} \frac{2}{x} - 2\lambda x &= 0, \\ \frac{5}{y} - 2\lambda y &= 0, \\ \frac{1}{z} - \lambda &= 0, \\ x^2 + y^2 + z &= 32. \end{aligned}$$

Tutorial Q3(a): Solve the conditions

Solution

The stationarity equations give

$$x^2 = \frac{1}{\lambda}, \qquad y^2 = \frac{5}{2\lambda}, \qquad z = \frac{1}{\lambda}.$$

Inserting them into $x^2 + y^2 + z = 32$ yields

$$\frac{1}{\lambda} + \frac{5}{2\lambda} + \frac{1}{\lambda} = 32 \quad \Longrightarrow \quad \lambda^* = \frac{9}{64}.$$

Since $x, y, z > 0$, the unique point satisfying the Lagrange conditions is

$$\boxed{(x^*, y^*, z^*, \lambda^*) = \left(\frac{8}{3}, \frac{4\sqrt{10}}{3}, \frac{64}{9}, \frac{9}{64}\right).}$$

Tutorial Q3(b): Approximate optimal value

Solution

Write $V(b)$ for the optimal value when the right-hand side of the equality constraint is b.
The optimal value for the original problem (as assumed in part (b)) is

$$V(32) = 2\ln\left(\frac{8}{3}\right) + 5\ln\left(\frac{4\sqrt{10}}{3}\right) + \ln\left(\frac{64}{9}\right).$$

The right-hand side increases by $\Delta b = 0.1$. Using the multiplier $\lambda^* = 9/64$ from part (a),

$$\begin{aligned} V(32.1) &\approx V(32) + \lambda^* \Delta b \\ &= V(32) + \frac{9}{64} \cdot \frac{1}{10}. \end{aligned}$$

Therefore,

$$\boxed{V(32.1) \approx V(32) + \frac{9}{640}.}$$

This is a first-order approximation, not an exact solution of the modified maximization problem.